ULTRASONIC SENSOR EXERCISE

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Explain and demonstrate how to acquire distance measurements using a digital ultrasonic sensor.	☐
2	Design and implement a Last Good Value (LGV) logic filter to manage erroneous or intermittent readings from a digital sensor.	☐

2.0 Hardware Set-Up

The exercise involves connecting an HC-SR04 ultrasonic sensor to an Arduino Uno to measure distance, as shown in Figure 1.

Note: While the ACE-Box (Base + Sense) can be used for this exercise, it is not required. Only the individual components listed below are needed.

Required hardware

- ATmega328P microcontroller (Arduino Uno)
- USB Type-B cable
- Breadboard
- HC-SR04 ultrasonic sensor
- 4 × male-to-male breadboard wires

HC-SR04 Ultrasonic Sensor Specifications

- Distance measurement range: 0.02 to 4.00 m
- Accuracy: ±0.003 m

Hardware Assembly Steps

1. Insert the ultrasonic sensor into the breadboard, as shown in Figure 1.

2. Connect the sensor's Trig pin to Pin 5 and its Echo pin to Pin 4 on the Arduino.
3. Connect the VCC terminal of the sensor to the 5 V supply pin on the Arduino Uno.
4. Connect the GND terminal of the sensor to a GND pin on the Arduino Uno.

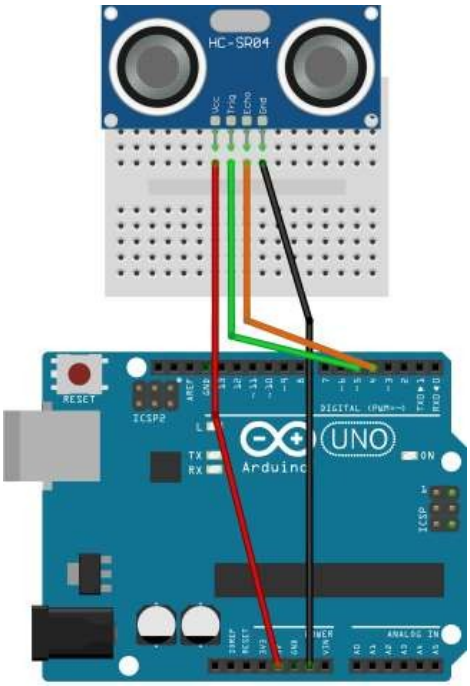


Figure 1: *Hardware Set-Up for the Ultrasonic Sensor Exercise*

3.0 Simulink Set-Up and Distance Measurement Testing

The complete Simulink model for the hardware described above is shown in Figure 2. The Ultrasonic Sensor block is found under Sensors in the Simulink Support Package for Arduino Hardware. The Display and Scope blocks are found under Sinks in Simulink.

Simulink Model Construction Steps

1. Open the Ultrasonic Sensor block parameters and set the number of signal pins to 2.
2. Set the Trigger pin to 5 and the Echo pin to 4, matching the wiring in Section 2.0.
3. Select a sample time (e.g., 0.01 s). Experiment with different values to determine the most suitable setting.
4. Run the Simulink model and observe the measured distance to the object ahead on the Scope.

To recap how to run Simulink code generation and deploy the model to the Arduino Uno, see Section 4.0 of the LED (Digital) Exercise or visit Code Generation.

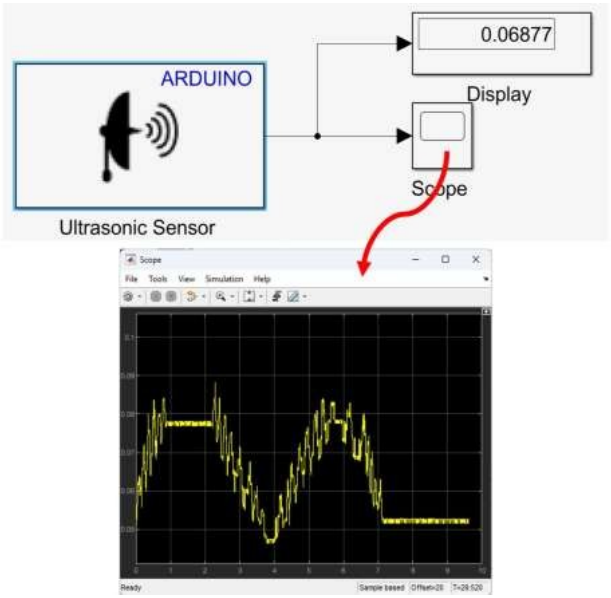


Figure 2: *Simulink Set-Up for the Ultrasonic Sensor Exercise*

4.0 Last Good Value (LGV) Logic Filter and Testing

A last good value (LGV) logic filter maintains a valid output when the input signal becomes unreliable or invalid. It is particularly useful where a sensor may occasionally fail or produce noisy, erroneous data. The filter "remembers" the last valid input value and continues to output it until a new, valid reading is received. For example, if a parking-assistance sensor on a vehicle temporarily fails or goes out of range, an LGV filter ensures the system still provides useful information based on the last good reading.

The filter can be configured to suit the operating range of the ultrasonic sensor (0.02 to 4.00 m). It is implemented in Simulink using the following blocks:

- **Switch** (Simulink → Signal Routing): controls the flow of data based on the sensor's output.
- **Memory** (Simulink → Discrete): stores the last valid reading, which is used when the sensor output falls below the threshold.

LGV Filter Set-Up Steps

1. Connect the blocks as shown in Figure 3.
2. Open the Switch block parameters and set the criterion to pass the first input when it is greater than a Threshold of 0.1.
3. Add a Mux and Scope blocks so that the signals can be viewed in real time.
4. Deploy the model and test its operation by placing an object within 0.1 m of the sensor and then beyond this distance.

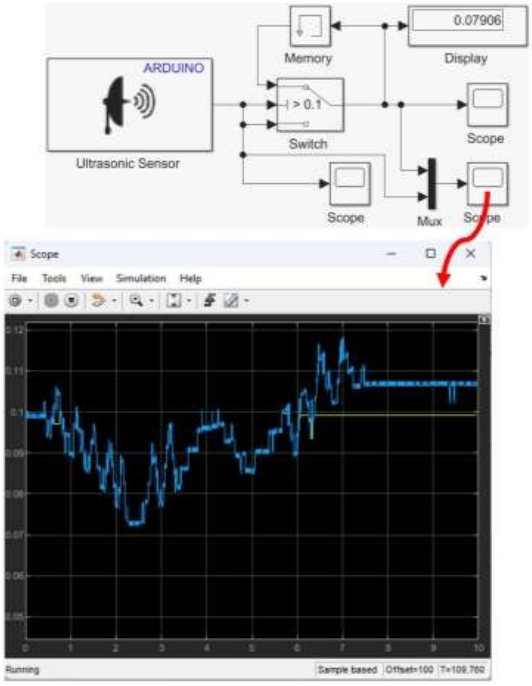


Figure 3: *Last Good Value (LGV) Logic Filter Simulink Model and Testing*

5.0 Exercises

1. **Sample time investigation:** test at least three different sample times and comment on the performance of the measurements.
2. **Static accuracy check:** measure three known distances and compare the readings against the reference values.
3. **Improved validity logic:** replace the 0.1 m threshold with a full range check (0.02 m to 4.00 m) and compare the behaviour.
4. **Moving target test:** move an object towards and away from the sensor and observe the smoothness and delay of the signal.

6.0 Concluding Remarks

This exercise has demonstrated practical distance measurement using an ultrasonic sensor integrated with Simulink and Arduino hardware. It reinforces correct hardware configuration, real-time data acquisition and the validation of sensor signals.

The implementation of the last good value (LGV) filter highlights the importance of basic signal conditioning to improve measurement reliability. These principles form the foundation for closed-loop control systems and for autonomous sensing applications such as parking aids.

Before completing the exercise, return to the Learning Outcomes in Table 1 and confirm that you are confident you can demonstrate each one. If so, tick the corresponding Achieved box.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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